

Änderung ergibt Trigger	Prio 1	Prio 2	Prio 3
Änderung erwartet, kein Trigger			
Multiple Instanzen			
Message Type dependent			

AS-2011 AS-2020

01 - list
res - 2

reserved
octet 1 bit 0-5

reserved
All 0's

0x05
0x00

PTP Messages 10.6

7	6	5	4	3	2	1	0	Octets	Offset
gPTP Message Header									34
majorSdoId		messageType						1	0
minorVersionPTP		versionPTP						1	1
		messageLength						2	2
		domainNumber						1	4
		minorSdoId						1	5
		flags						2	6
		correctionField						8	8
		messageTypeSpecific						4	16
		sourcePortIdentity						10	20
		sequenceId						2	30
		controlField						1	32
		logMessageInterval						1	33

7	6	5	4	3	2	1	0	Octets	Offset
Announce Message: 68 + N*8									68 + 8 * (stepsRemoved+1)
		header (see 10.6.2)						34	0
		reserved						10	34
		currentUtcOffset						2	44
		reserved						1	46
		grandmasterPriority1						1	47
		grandmasterClockQuality						4	48
		grandmasterPriority2						1	52
		grandmasterIdentity						8	53
		stepsRemoved						2	61
		timeSource						1	63
		path trace TLV: tvType 0x08						2	64
		path trace TLV: lengthField						2	66
		path trace TLV: pathSequence						N * 8	68

7	6	5	4	3	2	1	0	Octets	Offset
Signaling Message									60
		header (see 10.6.2)						34	0
		targetPortIdentity = all ones						10	34
		one of: message interval request TLV, gPTP-capable TLV, or gPTP-capable message interval request TLV						16	44

7	6	5	4	3	2	1	0	Octets	Offset
1-Step Sync									76
		header (see 11.4.2)						34	0
		originTimestamp						10	34
		Follow_Up information TLV: tvType						2	44
		Follow_Up information TLV: lengthField						2	46
		organizationId						3	48
		organizationSubType						3	51
		cumulativeScaledRateOffset						4	54
		gmTimeBaseIndicator						2	58
		lastGmPhaseChange						12	60
		scaledLastGmFreqChange						4	72

7	6	5	4	3	2	1	0	Octets	Offset
2-Step Sync									44
		header (see 11.4.2)						34	0
		reserved						10	34

7	6	5	4	3	2	1	0	Octets	Offset
Follow-Up									76
		header (see 11.4.2)						34	0
		preciseOriginTimestamp						10	34
		Follow_Up information TLV: tvType = 0x03						2	44
		lengthField = 28						2	46
		organizationId = 00-80-C2						3	48
		organizationSubType = 1						3	51
		cumulativeScaledRateOffset						4	54
		gmTimeBaseIndicator						2	58
		lastGmPhaseChange						12	60
		scaledLastGmFreqChange						4	72
		Drift_Tracking TLV: tvType = 0x03						2	76
		lengthField = 32						2	78
		organizationId = 00-80-C2						3	80
		organizationSubType = 6						3	83
		syncGressTimestamp						12	86
		syncGrandmasterIdentity						8	98
		syncStepsRemoved						2	106
		rateRatioDrift						4	108

7	6	5	4	3	2	1	0	Octets	Offset
Pdelay_Req Message									54
		header (see 11.4.2)						34	0
		reserved						10	34
		reserved						10	44

7	6	5	4	3	2	1	0	Octets	Offset
Pdelay_Resp Message									54
		header (see 11.4.2)						34	0
		requestReceiptTimestamp						10	34
		requestingPortIdentity						10	44

7	6	5	4	3	2	1	0	Octets	Offset
Pdelay_Resp_Follow_Up Message									54
		header (see 11.4.2)						34	0
		responseOriginTimestamp						10	34
		requestingPortIdentity						10	44

Msg type	Msg class	Value
Sync	Event	0x0
Pdelay_Req	Event	0x2
Pdelay_Resp	Event	0x3
Follow_Up	General	0x8
Pdelay_Resp_Follow_Up	General	0xA
Announce	General	0xB
Signaling	General	0xC

Table 10-9—Values of flag bits

Octet	Bit	Message types	Name	Value
0	0	All	alternateMasterFlag in Announce, Follow_Up, and Delay_Resp messages	Not used in this standard; transmitted as FALSE and ignored on reception
0	1	Sync, Pdelay_Req	trackingFlag	For Sync messages: a) For a one-step transmitting PTP Port (see 11.1.3 and 11.2.11.9), the value is FALSE. b) For a two-step transmitting PTP Port, the value is TRUE. For Pdelay_Req messages: The value is transmitted as TRUE and ignored on reception.
0	2	All	successFlag	Not used in this standard; transmitted as FALSE and ignored on reception
0	3	All	Reserved	Not used by IEEE Std 1588-2019; reserved as FALSE and ignored on reception
0	4	All	Reserved	Not used by IEEE Std 1588-2019; reserved as FALSE and ignored on reception
0	5	All	PTP profile specific 1	Not used in this standard; transmitted as FALSE and ignored on reception
0	6	All	PTP profile specific 2	Not used in this standard; transmitted as FALSE and ignored on reception
0	7	All	Reserved	Not used in this standard; transmitted as FALSE and ignored on reception
1	0	Announce	logp1	The value of the global variable logp1 (see 10.3.9.6)
1	1	Announce	logp5	The value of the global variable logp5 (see 10.3.9.5)
1	2	Announce	currentUtcOffsetValid	The value of the global variable currentUtcOffsetValid (see 10.3.9.6)
1	3	Announce	pppTraceable	The value of the global variable pppTraceable (see 10.3.9.7)

Table 10-9—Values of flag bits (continued)

Octet	Bit	Message types	Name	Value
1	4	Announce	timeTraceable	The value of the global variable timeTraceable (see 10.3.9.8)
1	5	Announce	frequencyTraceable	The value of the global variable frequencyTraceable (see 10.3.9.8)
1	6	All	Reserved	Not used by IEEE Std 1588-2019; reserved as FALSE and ignored on reception
1	7	All	Reserved	Not used in this standard; reserved as FALSE and ignored on reception

Signaling Messages:
Signaling Msg mit gPTP-capable TLV muss einfach zyklisch kommen.
Alle andere Signaling messages triggern eine Aufzeichnung